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Tie bar for shear wall, shear wall, and method for constructing the same

Abstract

A tie bar for fixing opposing steel meshes in a shear wall is provided. The shear wall includes a first steel mesh and a second steel mesh. The first steel mesh and the second steel mesh are spaced apart by a predetermined distance. The tie bar includes a first end portion, a second end portion, and two connections. The first end portion is configured to surround and be fixed to two adjacent vertical bars of the first steel mesh. The second end portion includes two free ends, and these two free ends are configured to respectively be fixed to two adjacent vertical bars of the second steel mesh. The two connections connect the first end portion and the second end portion respectively.

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Background/Summary

BACKGROUND

1. Technical Field

(1) The present disclosure relates to a tie bar for a shear wall, a shear wall including the tie bar, and a method for constructing the same.

2. Description of the Related Art

(2) Conventional shear wall construction normally includes providing opposing steel meshes comprising intersecting vertical steel bars, spacing the two meshes by a distance, fixing and connecting the two meshes with conventional tie bars, enclosing the meshes with formworks, and filling spaces in the formworks with concrete.

(3) Conventional tie bars generally consist of a straight steel bar having a hook at each of its respective ends. In a conventionally constructed shear wall, the hooks are fixed to the vertical bars of the two meshes at intervals of around four times the distance between the two meshes. Such arrangement requires use of a large number of tie bars for secure connection thereof, increasing

labor and costs.
(4) Therefore, it is desirable to provide a tie bar for a shear wall requiring fewer tie bars to be used.
SUMMARY
(5) In one aspect, a tie bar for fixing opposing steel meshes in a shear wall is provided. The shear wall comprises a first steel mesh and a second steel mesh. The first steel mesh and the second steel mesh are spaced apart by a predetermined distance. The tie bar includes a first end portion, a second end portion and two connections. The first end portion is configured to surround and be fixed to two adjacent upstanding vertical bars of the first steel mesh. The second end portion has two free ends which are configured to respectively be fixed to two adjacent upstanding vertical bars of the second steel mesh. The two connections connect the first end portion and the second end portion respectively
(6) In another aspect, a construction method for a shear wall is provided. The method comprises providing two precast columns spaced apart by a predetermined distance, providing a plurality of tension bars in the two precast columns, wherein each of the plurality of tension bars has nut heads at two ends for connecting to bars, the nut heads being exposed, providing and connecting extending bars to the corresponding nut heads of the tension bars, providing a first steel mesh and a second steel mesh and disposing them between the two precast columns, wherein the first steel mesh and the second steel mesh are spaced apart by a predetermined distance, connecting the lateral ends of the first steel mesh and the second steel mesh to the extending bars, and providing a plurality of the tie bars and using the same to fix the first steel mesh and the second steel mesh. Each of the tie bars is configured to surround two adjacent upstanding vertical bars of the first steel mesh and two adjacent upstanding vertical bars of the second steel mesh.
(7) The foregoing has outlined rather broadly the features and technical advantages of the present disclosure in order that the detailed description of the disclosure that follows may be better understood. Additional features and advantages of the disclosure will be described hereinafter, and form the subject of the claims of the disclosure. It should be appreciated by those skilled in the art that the conception and specific embodiment disclosed may be readily utilized as a basis for modifying or designing other structures or processes for carrying out the same purposes of the present disclosure. It should also be realized by those skilled in the art that such equivalent constructions do not depart from the spirit and scope of the disclosure as set forth in the appended claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) Aspects of some embodiments of the present disclosure are readily understood from the following detailed description when read with the accompanying figures. It should be noted that various structures may not be drawn to scale, and dimensions of the various structures may be arbitrarily increased or reduced for clarity of discussion.
(2) FIG. **1** is a schematic view of a tie bar for a shear wall in accordance with one embodiment of the present disclosure.
(3) FIG. **2**A is a first schematic view showing a construction method in accordance with an embodiment of the present disclosure.
(4) FIG. **2**B is a second schematic view showing the structures of the precast columns in accordance with the embodiment of the present disclosure.
(5) FIG. **2**C is a third schematic view showing the construction method in accordance with the embodiment of the present disclosure.
(6) FIG. **2**D is a fourth schematic view showing the construction method in accordance with the embodiment of the present disclosure.

(7) FIG. **2**E is a schematic view along an X**1** orientation as shown in FIG. **2**D.
(8) FIG. **2**F is a fifth schematic view showing the construction method in accordance with the embodiment of the present disclosure.
(9) FIG. **2**G is a sixth schematic view showing the construction method in accordance with the embodiment of the present disclosure.
(10) FIG. **2**H is a seventh schematic view of a construction method in accordance with the embodiment of the present disclosure.
(11) FIG. **2**I is a schematic view along an X**2** orientation as shown in FIG. **2**H.
(12) FIG. **2**J is an eighth schematic view showing the construction method in accordance with the embodiment of the present disclosure.
(13) FIG. **2**K is a ninth schematic view showing the construction method in accordance with the embodiment of the present disclosure.
(14) FIG. **3**A is a first schematic view showing a construction method in accordance with another embodiment of the present disclosure.
(15) FIG. **3**B is a second schematic view showing the construction method in accordance with said another embodiment of the present disclosure.
(16) FIG. **3**C is a third schematic view showing the construction method in accordance with said another embodiment of the present disclosure.
(17) FIG. **3**D is a fourth schematic view showing the construction method in accordance with said another embodiment of the present disclosure.
(18) FIG. **3**E is a schematic view from a perspective along a Y**1** orientation as shown in FIG. **3**D.
(19) FIG. **3**F is a fifth schematic view showing the construction method in accordance with said another embodiment of the present disclosure.
(20) FIG. **3**G is a sixth schematic view showing the construction method in accordance with said another embodiment of the present disclosure.
(21) FIG. **3**H is a seventh schematic view showing the construction method in accordance with said another embodiment of the present disclosure.
(22) FIG. **3**I is a schematic view from a perspective along a Y**2** orientation as shown in FIG. **3**H.
(23) FIG. **3**J is an eighth schematic view showing the construction method in accordance with said another embodiment of the present disclosure.
(24) FIG. **3**K is a ninth schematic view showing the construction method in accordance with said another embodiment of the present disclosure.

DETAILED DESCRIPTION

(25) The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.
(26) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted

accordingly.

(27) It should be understood that, although the terms first, second, etc. may be used herein to describe various elements, these elements should not be limited by these terms. Unless indicated otherwise, these terms are only used to distinguish one element from another element.

(28) As used herein, the terms “approximately,” “substantially,” “substantial” and “about” are used to describe and account for small variations. When used in conjunction with an event or circumstance, the terms can refer to instances in which the event or circumstance occurs precisely as well as instances in which the event or circumstance occurs to a close approximation. For example, when used in conjunction with a numerical value, the terms can refer to a range of variation less than or equal to ±10% of that numerical value, such as less than or equal to ±5%, less than or equal to ±4%, less than or equal to ±3%, less than or equal to ±2%, less than or equal to ±1%, less than or equal to ±0.5%, less than or equal to ±0.1%, or less than or equal to ±0.05%. For example, two numerical values can be deemed to be “substantially” the same or equal if a difference between the values is less than or equal to ±10% of an average of the values, such as less than or equal to ±5%, less than or equal to ±4%, less than or equal to ±3%, less than or equal to ±2%, less than or equal to ±1%, less than or equal to ±0.5%, less than or equal to ±0.1%, or less than or equal to ±0.05%.

(29) FIG. **1** is a schematic view of a tie bar **1** for a shear wall in accordance with one embodiment of the present disclosure. As shown in FIG. **1**, the tie bar **1** of the present disclosure includes a first end portion **11**, a second end portion **12**, and two connections **13** connecting the first end portion **11** and the second end portion **12** respectively. The first end portion **11**, the second portion **12** and the connections **13** are formed in one piece. The first end portion **11** is substantially inverted U-shaped and includes two bending sections **111** at its two sides. The second end portion **12** includes two free ends, each free end having a hook **121**. In some embodiments of the present disclosure, a distance between the two bending sections **111** of the first end portion **11** is substantially equal to a distance between the two hooks **121** of the second end portion **12**. In some embodiments of the present disclosure, one end of each of the connections **13** connects the bending section **111** of the first end portion **11** and the other end of each of the connections **13** connects the hook **121** of the second end portion **12**. A distance between the two connections **13** gradually decreases along a direction from the first end portion **11** toward the second end portion **12**. Further, the two hooks **121** of the second end portion **12** are opposite to each other and bent toward the outside of the tie bar **1**.

(30) FIGS. **2**A-**2**K are a series of schematic views illustrating a construction method for manufacturing shear walls **100** and **100′** by using the tie bar **1** according to an embodiment of the present disclosure.

(31) As the embodiment shown in FIG. **2**A, the construction method provides several precast columns **31**, **32**, **33** on a building site, the precast columns spaced apart by a predetermined distance. In some embodiments of the present disclosure, the distance between the precast columns **31** and the precast columns **32** is greater than, equal to, or less than a distance between the precast columns **32** and the precast columns **33**. A plurality of vertical steel bars **37** are arranged on the building site and between the pillars **31**, **32** and **33** for connecting and fixing the bottom sides of the steel mesh **10** and the steel mesh **20** (see FIGS. **2**C and **2**D).

(32) Please refer to FIG. **2**B, which shows the structures of the precast columns **31**, **32**, **33**. Each of the precast columns **31**, **32**, **33** is provided with a plurality of tension bars **39** along both the height and width of the precast columns **31**, **32**, **33**. The end of each tension bar **39** is provided with nut heads **391**, configured to connect steel bars. A plurality of extending bars **311** are respectively connected with the nut heads **391** of the corresponding tension bars **39**. The extending bars **311** serve as extensions to be connected with the two sides of the steel mesh **10** and the steel mesh **20** (see FIGS. **2**C and **2**D). In some embodiments of the present disclosure, at least one of the extending bars **311** at two sides of the precast column **31** extends toward the precast column **32**, and at least one of the extending bars **321** at two sides of the precast column **32** extends toward the

precast column **31**, and at least one of the extending bars **321** at two sides of the precast column **32** extends toward the precast column **33**, and at least one of the extending bars **331** at two sides of the precast column **33** extends toward the precast column **32**. The extending bars **311**, **321**, **331** may be threaded or a threaded at its end.

(33) As shown in FIG. **2**C, the steel meshes **10**, **20** are disposed between the precast column **31** and the precast column **32**. Each of the precast columns **31**, **32**, **33** at each side has two parallel rows of extending bars **311**, **321**, **331** disposed along the height thereof. The steel mesh **10** and the steel mesh **20** are spaced apart by a predetermined distance, and two sides of the steel mesh **10** are connected to a row of extending bars **311** of the precast column **31** and a row of extending bars **321** of the precast **32** respectively and two sides of the steel mesh **20** are connected to the other row of the extending bars **31** of the precast column **31** and the other row of extending bars **321** of the precast **32** respectively. Moreover, the steel meshes **10**′, **20**′ are disposed between the precast column **32** and the precast column **33**. The steel mesh **10**′ and the steel mesh **20**′ are spaced apart by a predetermined distance, and two sides of the steel mesh **10**′ are connected to a row of extending bars **321** of the precast column **32** and a row of extending bars **331** of the precast column **33** respectively and two sides of the steel mesh **20**′ are connected to the other row of extending bars **321** of the precast column **32** and the other row of extending bars **331** of the precast **33** respectively.

(34) As shown in FIG. **2**D, the construction method provides one or more tie bars **1** as shown in FIG. **1** between the steel meshes **10** and **20**, and the tie bars **1** connect and fix the steel meshes **10** and **20**. Further, one or more tie bars **1** as shown in FIG. **1** are also arranged between the steel meshes **10**′ and **20**′, and the tie bars **1** to connect and fix the steel meshes **10**′ and **20**′.

(35) FIG. **2**E is a schematic view along an X**1** orientation as shown in FIG. **2**D. As shown in FIG. **2**E, the steel mesh **10** and the steel mesh **20** are opposite to each other and are spaced apart with a predetermined distance H. The steel mesh **10** includes a plurality of upstanding vertical bars **101** spaced apart by a distance S**1**. Further, the steel mesh **20** includes a plurality of upstanding vertical bars **201** spaced apart by a distance S**2**. In some embodiments of the present disclosure, the distance S**1** is substantially equal to the distance S**2**.

(36) Referring to FIG. **2**E again, the tie bars **1** as shown in FIG. **1** are arranged to connect the steel mesh **10** and the steel mesh **20**. Each of the tie bars **1** connects two adjacent upstanding vertical bars **101** of the steel mesh **10** and two adjacent upstanding vertical bars **201** of the steel mesh **20**. The substantially inverted U-shaped first end portion **11** of the tie bar **1** is configured to surround the outer circumferences of the two adjacent upstanding vertical bars **101** of the steel mesh **10**. The two bending sections **111** of the first end portion **11** respectively hold or retain the two adjacent upstanding vertical bars **101** of the steel mesh **10**. Further, the two hooks **121** of the second portion **12** of the tie bar **1** respectively hook the two adjacent upstanding vertical bars **201** of the steel mesh **20**. Moreover, the two connections **13** of the tie bar **1** respectively extend from outer sides of the two adjacent upstanding vertical bars **101** of the steel mesh **10** to the inner sides of the two adjacent upstanding vertical bars **201** of the steel mesh **20**. As shown in FIG. **2**E, an extending direction of the connection **13** of the tie bar **1** and the extending direction of the steel mesh **10** or the extending direction of the steel mesh **20** form a non-right angle.

(37) In the embodiment shown in FIG. **2**E, the orientations of each two adjacent tie bars **1** are the same, and each two adjacent tie bars **1** connect and secure the steel meshes **10** and **20** in the same way. For example, both of them are connected to the two adjacent upstanding vertical bars **101** of the metal mesh **10** with the first end portions **11**, and both of them are connected to the two adjacent vertical bars **201** of the metal mesh **20** with the second end portion S**12**.

(38) Moreover, each two adjacent tie bars **1** are spaced apart by a distance D**1** in the transverse direction in FIG. **2**E. In some embodiments of the present disclosure, the distance D**1** between the two adjacent tie bars is substantially equal to eight times the distance (S**1**) between two adjacent upstanding vertical bars **101** of the metal mesh **10** or substantially equal to eight times the distance

(S2) between two adjacent upstanding vertical bars **201** of the metal mesh **20**.

(39) Further, in addition to being arranged in the transverse direction in FIG. **2**E, some of the plurality of tie bars **1** (not shown) are arranged lengthwise along the upstanding vertical bars **101** of the metal mesh **10** or lengthwise along the upstanding vertical bars **201** of the metal mesh **20**. In some embodiments of the present disclosure, another tie bar **1** (not shown), the same as or similar to the tie bar **1** as shown in FIG. **1**, is arranged underneath a tie bar **1** shown in FIG. **2**E by another predetermined distance, and the substantially inverted U-shaped first end portion of the lower tie bar is configured to surround the two adjacent upstanding vertical bars **101** of the metal mesh **10** and the two hooks of the second end portion of the lower tie bar respectively hook the two adjacent upstanding vertical bars **201** of the metal mesh **20**. That is, the orientations of the upper tie bar and the lower tie bar are the same, and the upper tie bar and the lower tie bar connect the steel meshes **10**, **20** in the same way.

(40) As shown in FIG. **2**F, a precast beam **301** is disposed above the steel mesh **10** and the steel mesh **20** and extends between the upper end of the precast column **31** and the upper end of the precast column **32**. The precast beam **301** at its lower side has extending bars **3011** for connection with the steel meshes **10**, **20**, and there is a distance of around 5 cm (not shown) between the lower surface of the precast beam **301** and the top of the steel mesh **10** or that of the steel mesh **20**. Moreover, a precast beam **302** is disposed above the steel mesh **10′** and the steel mesh **20′** and extends between the upper end of the precast column **32** and the upper end of the precast column **33**. The precast beam **302** at its lower side has extending bars **3021** for connection with the steel meshes **10′**, **20′**, and there is a distance of around 5 cm (not shown) between the lower surface of the precast beam **302** and the top of the steel mesh **10′** or that of the steel mesh **20′**. In some embodiments of the present disclosure, the precast beam **301**, **302** is a fully prefabricated beam or a semi-precast beam.

(41) As shown in FIGS. **2**G, **2**H, the construction method of the present disclosure provides formworks **41**, **42**, **41′**, **42′**. The formworks **41** and **42** are disposed so that the steel meshes **10** and **20** are received therebetween. In other words, the steel meshes **10** and **20** are disposed between the formworks **41** and **42**. The formworks **41′** and **42′** are disposed so that the steel meshes **10′** and **20′** are received therebetween. In other words, the steel meshes **10′** and **20′** are disposed between the formworks **41′** and **42′**.

(42) FIG. **2**I is a schematic view along an X**2** orientation as shown in FIG. **2**H. As shown in FIG. **2**I, the steel meshes **10** and **20** are parallel to each other and are disposed between the precast columns **31** and **32**. The two sides of steel meshes **10** and **20** are connected to the extending bars **311** of the precast column **31** and the extending bars **321** of the precast column **32**. A plurality of tie bars **1** connect and secure the steel meshes **10** and **20** in the manner as shown in FIG. **2**D. As shown in FIG. **2**I, the formworks **41** and **42** are configured to receive the steel meshes **10** and **20** therebetween. Further, a plurality of pulling sheets **45** connect the formworks **41** and **42**. Each of the pulling sheets **45** is configured to pass through the formwork **41** and the formwork **42**, and its two ends are fixed to the formwork **41** and the formwork **42** thereby tightening the formwork **41** and the formwork **42**.

(43) As shown in FIG. **2**J, the formwork **41** has outlets **411**, **413** adjacent to its upper end and the formworks **42** has inlet **421** and outlet **423** adjacent to its upper end. The concrete **50** is poured between the formworks **41** and **42** and between formworks **41′** and **42′** (not shown). Taking the embodiment shown in FIG. **2**J as an example, the concrete **50** is poured into the space between the formworks **41** and **42** through the inlet **421** of the formwork **42**. When the concrete **50** is poured into the space between the formworks **41** and **42** until reaching a level close to the upper ends of the formworks **41** and **42**, air and a small amount of excess concrete **50** is expelled from the outlets **411**, **413**, **423**. When the small amount of the concrete **50** is expelled from the outlets **411**, **413**, **423**, it signifies that the air has been removed and the concrete pouring can stop. In addition, after the initial setting of the concrete **50**, expansive-cement mortar is poured into a space (not shown)

between the lower surface of the precast beam **301** and the tops of the steel mesh **10**, **20** and a space (not shown) between the lower surface of the precast beam **302** and the tops of the steel mesh **10′**, **20′**, which is around 5 cm in height. FIG. **2**K shows a shear wall **100** and a shear wall **100′** which are formed after the concrete **50** is solidified and formed and the formworks **41**, **42** and the formworks **41′** and **42′** in FIG. **2**G are removed.

(44) FIGS. **3**A-**3**K are a series of schematic views illustrating a construction method for manufacturing shear walls **200** and **200′** by using the tie bar **1** according to another embodiment of the present disclosure.

(45) As shown in FIG. **3**A, the construction method of the another embodiment provides several precast columns **34**, **35**, **36** on a building site, and the precast columns **34**, **35**, **36** are spaced apart by a predetermined distance. In some embodiments of the present disclosure, the distance between the precast columns **34** and the precast columns **35** is greater than, equal to, or less than the distance between the precast columns **35** and the precast columns **36**. A plurality of vertical steel bars **38** are arranged on the building site and between the pillars **34**, **35** and **36** for connecting with and fixing to the bottom sides of the steel mesh **60** and the steel mesh **70** (see FIGS. **3**C and **3**D).

(46) Referring to FIG. **3**B, each of the precast columns **34**, **35**, **36** is provided with a plurality of tension bars **39′** disposed along the height and width (not shown) of the precast columns **34**, **35**, **36**. Each of the tension bars **39′** is provided with nut heads **391′**, configured to connect bars. A plurality of extending bars **341** are respectively connected with the nut heads **391′** of the corresponding tension bars **39′** as extensions to connect with the two sides of the steel mesh **60** and the steel mesh **70** (see FIGS. **3**C and **3**D). In some embodiments of the present disclosure, at least one of the extending bars **341** at two sides of the precast column **31** extends toward the precast column **35**, and at least one of the extending bars **351** at two sides of the precast column **35** extends toward the precast column **34**, and at least one of the extending bars **351** at two sides of the precast column **35** extends toward the precast column **36**, and at least one of the extending bars **361** at two sides of the precast column **36** extends toward the precast column **35**. The extending bars **341**, **351**, **361** may be threaded or threaded at the ends.

(47) As shown in FIG. **3**C, the steel meshes **60**, **70** are disposed between the precast column **34** and the precast column **35**. The steel mesh **60** and the steel mesh **70** are spaced apart by a predetermined distance, and two sides of the steel mesh **60** are connected to the extending bars **341** of the precast column **34** and the extending bars **351** of the precast **35** respectively and two sides of the steel mesh **70** are connected to the other extending bars **34** of the precast column **34** and the other extending bars **351** of the precast **35** respectively. Moreover, the steel meshes **60′**, **70′** are disposed between the precast column **35** and the precast column **36**. The steel mesh **60′** and the steel mesh **70′** are spaced apart by a predetermined distance, and two sides of the steel mesh **60′** are connected to the extending bars **351** of the precast column **35** and the extending bars **361** of the precast column **36** respectively and two sides of the steel mesh **70′** are connected to the other extending bars **351** of the precast column **35** and the other extending bars **361** of the precast **36** respectively.

(48) As shown in FIG. **3**D, the construction method provides one or more tie bars **1** as shown in FIG. **1** between the steel meshes **10** and **20**, and the tie bars **1** are connected with and fixed to the steel meshes **10** and **20**. Further, one or more tie bars **1** as shown in FIG. **1** are also arranged to be disposed between the steel meshes **10′** and **20′** for connection with and fixing to the steel meshes **10′** and **20′**.

(49) FIG. **3**E is a schematic view along a Y**1** orientation as shown in FIG. **3**D. As shown in FIG. **3**E, the steel mesh **60** and the steel mesh **70** are opposite to each other and spaced apart by a predetermined distance. The steel mesh **60** includes a plurality of upstanding vertical bars **601** spaced apart by a distance S**3**. Further, the steel mesh **70** includes a plurality of upstanding vertical bars **701** spaced apart by a distance S**4**. In some embodiments of the present disclosure, the distance S**3** is substantially equal to the distance S**4**.

(50) Please refer to FIG. **3**E. The tie bars **1** as shown in FIG. **1** are configured to connect the steel mesh **60** and the steel mesh **70**. The tie bars **1** shown in FIG. **3**E connect the two adjacent upstanding vertical bars **601** of the steel mesh **60** and the two adjacent upstanding vertical bars **701** of the steel mesh **70** such that the steel meshes **60** and **70** are connected and secured to each other. The substantially inverted U-shaped first end portion **11** of the tie bar **1** is configured to surround the outer circumferences of e two adjacent upstanding vertical bars **601** of the steel mesh **60**. The two bending sections **111** of the first end portion **11** may respectively hold or retain two adjacent upstanding vertical bars **601** of the steel mesh **60**. Further, the two hooks **121** of the second portion **12** of the tie bar **1** respectively hook two adjacent upstanding vertical bars **701** of the steel mesh **70**. Moreover, the two connections **13** of the tie bar **1** respectively extend from the outer sides of the two adjacent upstanding vertical bars **601** of the steel mesh **60** to the inner sides of two adjacent upstanding vertical bars **701** of the steel mesh **70**. As shown in FIG. **3**E, the extending direction of the connection **13** of the tie bar **1** and the extending direction of the steel mesh **60** and/or the steel mesh **60** form a non-right angle.
(51) As shown in FIG. **3**E, the central tie bar **1** between the left side tie bar **1** and the right side tie bar **1** has an orientation different from that of the left side tie bar **1** and that of the right side tie bar **1**. The substantially inverted U-shaped first end portion **11** of the central tie bar **1** is configured to surround the outer circumferences of two adjacent upstanding vertical bars **701** of the steel mesh **70**. The two bending sections **111** of the first end portion **11** respectively hold or retain two adjacent upstanding vertical bars **701** of the steel mesh **70**. Further, the two hooks **121** of the second portion **12** of the central tie bar **1** respectively hook two adjacent upstanding vertical bars **601** of the steel mesh **60**. Furthermore, the two connections **13** of the central tie bar **1** respectively extend from outer sides of two adjacent upstanding vertical bars **701** of the steel mesh **70** to the inner sides of the two adjacent upstanding vertical bars **601** of the steel mesh **60**. As shown in FIG. **3**E, the extending direction of the connection **13** of the tie bar **1** and the extending direction of the steel mesh **60** and/or the steel mesh **60** form a non-right angle. In other words, in this embodiment, two adjacent tie bars **1** have opposite orientations. For example, when a tie bar **1** connects the two adjacent upstanding vertical bars **601** of the steel mesh **60** with its first end portion **11** and connects the two adjacent upstanding vertical bars **701** of the steel mesh **70** with the second end portion **12**, the other tie bar **1** connects the two adjacent upstanding vertical bars **701** of the steel mesh **70** with its first end portion **11** and connects the two adjacent upstanding vertical bars **601** of the steel mesh **60** with the second end portion **12**
(52) Moreover, two adjacent tie bars **1** are spaced apart by a distance D**2** in the transverse direction in FIG. **3**E. In some embodiments of the present disclosure, the distance between the two adjacent tie bars is substantially equal to eight times the distance S**3** between the two adjacent upstanding vertical bars **601** of the metal mesh **60** or substantially equal to eight times the distance S**4** between the two adjacent upstanding vertical bars **801** of the metal mesh **80**.
(53) Further, in addition to being disposed in the transverse direction in FIG. **3**E, some of the plurality of tie bars **1** (not shown) are arranged lengthwise along the upstanding vertical bar **601** of the metal mesh **60** or lengthwise along the upstanding vertical bar **701** of the metal mesh **70**. In some embodiments of the present disclosure, another tie bar **1** (not shown), which is the same as or similar to the tie bar **1** as shown in FIG. **1**, is disposed underneath a tie bar **1** shown in FIG. **2**E by another predetermined distance. Moreover, the structures and the orientations of the upper tie bar **1** and the lower tie bar **1** may be the same. In some embodiments of the present disclosure, the structures and the orientations of the upper tie bar **1** and the lower tie bar **1** are different.
(54) As shown in FIG. **3**F, a precast beam **305** is disposed above the steel mesh **60** and the steel mesh **70** and extends between the upper end of the precast column **34** and the upper end of the precast column **35**. The precast beam **305** at its lower side has extending bars **3051** for connection with the steel meshes **60**, and there is a distance of around 5 cm (not shown) between the lower surface of the precast beam **305** and the top of the steel mesh **60** or the steel mesh **70**. Moreover, a

precast beam **306** is disposed above the steel mesh **60′** and the steel mesh **70′** and extends between the upper end of the precast column **35** and the upper end of the precast column **36**. The precast beam **306** at its lower side has extending bars **3061** for connection to the steel meshes **60′**, **70′**, and there is a distance of around 5 cm (not shown) between the lower surface of the precast beam **306** and the top of the steel mesh **60′** or the steel mesh **70′**. In some embodiments of the present disclosure, the precast beam **305**, **306** is a fully prefabricated beam or a semi-precast beam.
(55) As shown in FIGS. **3**G, **3**H, the construction method of the present disclosure provides formworks **81**, **82**, **81′**, **82′**. The formworks **81** and **82** are disposed so that the steel meshes **60** and **70** are received therebetween. In other words, the steel meshes **60** and **70** are disposed between the formworks **81** and **82**. The formworks **81′** and **82′** are disposed so that the steel meshes **60′** and **70′** are received therebetween. In other words, the steel meshes **60′** and **70′** are disposed between the formworks **81′** and **82′**.
(56) FIG. **3**I is a schematic view from a perspective along a Y**2** orientation as shown in FIG. **3**H. As shown in FIG. **3**I, the steel meshes **60** and **70** are parallel to each other and are disposed between the precast columns **34** and **35**, and two sides of the steel meshes **60** and **70** are connected to the extending bars **341** of the precast column **34** and the extending bars **351** of the precast column **35**. A plurality of tie bars **1** connect the steel meshes **60** and **70** in the manner as shown in FIG. **3**D. As shown in FIG. **3**I, the formworks **81** and **82** are configured to receive the steel meshes **60** and **70** therebetween. Further, a plurality of pulling sheets **85** connect the formworks **81** and **82**. The pulling sheet **85** is configured to pass through the formwork **81** and the formwork **82**, and its two ends are fixed to the formwork **81** and the formwork **82** thereby tightening the formwork **81** and the formwork **82**.
(57) As shown in FIG. **3**J, the formwork **81** has outlets **811**, **813** adjacent to its upper end and the formworks **82** has an inlet **821** and an outlet **823** adjacent to its upper end. The concrete **90** is poured into the space between the formworks **81** and **82** and between formworks **81′** and **82′**. Taking the embodiment shown in FIG. **3**J as an example, the concrete **90** is poured into between space between the formworks **81** and **82** through the inlet **821** of the formwork **82**. When the concrete **90** is poured into the space between the formworks **81** and **82** until reaching a level close to the upper ends of the formworks **81** and **82**, the air and a small amount of excess concrete **90** are expelled from the outlets **811**, **813**, **823**. When the small amount of the concrete **90** is expelled from the outlets **811**, **813**, **823**, it means that the air has been removed and the concrete pouring can be stopped. In addition, after the initial setting of the concrete **90**, expansive-cement mortar (not shown) is poured into the space between the lower surface of the precast beam **305** and the tops of the steel mesh **60**, **70** and the space between the lower surface of the precast beam **306** and the tops of the steel mesh **60′**, **70′**, which is around 5 cm in height. FIG. **3**K show a shear wall **200** and a shear wall **200′** which are formed after the concrete **90** is solidified and formed and the formworks **81**, **82** and the formworks **81′** and **82′** shown in FIG. **3**G are removed.
(58) With the tie bar **1** of the present disclosure, the number of the tie bars used in manufacturing the same shear walls **100**, **100′**, **200**, **200′** can be greatly reduced, reducing costs and labor. The following Table 1 compares the number of bars required for a single shear wall using conventional tie bars and that using the tie bars of the present invention. As shown, the amount can be reduced by 7% to 8% if the tie bars of the present disclosure are used.
(59) TABLE-US-00001 TABLE 1 Conventional Tie Bar of the Tie Bar Present Disclosure Length (m)/per tie bar 0.959 1.773 Number of tie bars used in the 12 6 transverse direction Number of tie bars used in the 11.4 11.4 vertical direction Total number 136.5 68.25 Total length (m) 130.9 121.0 Total weight (kg) 130.1 120.3 Usage percentage 100% 92.4%
(60) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages

of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A construction method comprising: providing two precast columns spaced apart by a third predetermined distance; providing a plurality of tension bars in the two precast columns, wherein each of the plurality of tension bars has nut heads at two ends for connecting bars, the nut heads being exposed; providing and connecting extending bars to the corresponding nut heads of the tension bars; providing a first steel mesh and a second steel mesh and disposing them between the two precast columns, wherein the first steel mesh and the second steel mesh are spaced apart by a predetermined distance; connecting the lateral ends of the first steel mesh and the second steel mesh to the extending bars, and providing a plurality of tie bars to fix the first steel mesh and the second steel mesh; wherein each of the plurality of tie bars comprises: a first end portion configured to surround and be fixed to two adjacent upstanding vertical bars of the first steel mesh; a second end portion comprising two free ends, wherein the two free ends are configured to respectively be fixed to two adjacent upstanding vertical bars of the second steel mesh; and two connections connecting the first end portion and the second end portion respectively; wherein the first end portion, the second end portion and the two connections are integrally formed, and wherein the connections are non-perpendicular to the first steel mesh or to the second steel mesh; wherein the two free ends of the second end portion have two hooks respectively, and wherein the two hooks are respectively connected to the two connections and configured to respectively hook and to be fixed to the two adjacent upstanding vertical bars of the second steel mesh; wherein the two hooks are bent outwards; and wherein the two connections respectively extend from outsides of the two adjacent upstanding vertical bars of the first steel mesh to insides of the two adjacent upstanding vertical bars of the second steel mesh.
2. The construction method of claim 1, further comprising: providing two formworks, wherein the first steel mesh and the second steel mesh are arranged between the two formworks; providing a plurality of tension tabs, wherein two ends of the tension tabs are connected to the two formworks respectively; providing and hoisting a precast beam or a semi-precast beam to above the tops of the two precast column; fixing the two ends of the precast beam or a semi-precast beam to the tops of the two precast column respectively so that a spacing between a bottom surface of the precast beam or the semi-precast beam and the first steel mesh and the second steel mesh is about 5 cm; pouring concrete into between the two formworks; and pouring expansive-cement mortar into the spacing after the initial setting of the concrete.